

Towards a Readability Index for Music (R I M)

A cognitive-based theoretical approach to complexity in written music

Abstract

The present text describes the Indicators of a Readability Index for Music: the RIM. These indicators are the result of a model, built based on cognitive and mathematical concepts that assess the complexity of written music. Understanding readability as the ease with which we read a text, the RIM is designed to improve the perception of this feature in written music. Its construction is based on recent literature from music cognition, and it evaluates the syntactic complexity of a musical fragment using tools from Information Theory, such as Shannon's entropy. The model provides five indicators of complexity in written music. After computer testing and a case study of a fragment of Beethoven's *Largo* of the *3rd. Piano Concerto*, we conclude that these indicators reflect difficulty features in western music; according to the music cognition literature. These show minimal interdependence, as reported in statistical analysis in arts. The RIM is useful for music educators that can now compare complexity in written music, and assess its difficulty more homogeneously.

Key Words: Readability, Cognitive Musicology, Music Notation, Readability Formulas.

1 Introduction

Readability is an understudied area in music. Its assessment is often left to music editors, comments from other musicians, or fast glances at what is written. We understand readability as the ease with which we read a text (Bailin and Grafstein 2016; Benjamin 2012; Chitalkina et al. 2020; Jensen 2016; Matsubara et al. 2009; Sancho Guinda 2002; Sinha et al. 2019; Stenberg and Cross 2019; Tarasov 2015; Zhou et al. 2016). In music studies, we encounter at least three forms of music reading (i.e., three ways of transcoding visual symbols): 1. Playing-like as visual to motor transcoding; 2. Singing-like as visual to auditory transcoding; and 3. Naming-like notes as visual to verbal transcoding (Janurik et al. 2022: 2). So the question arises: In which of these three a musician builds a readability criterion? Whichever the option, this quality of a music text is a key concept that allows fluidity in all types of readings.

Unlike the above, quantitative research in music is an ongoing field. We propose a formal analysis of readability using as background readability formulas for literary texts, translated to music taking closely into account cognition research in the field of music reading. For this, we need to understand readability in a wider, and rigorous manner: Readability will be conceptualized as a bridge or a coded mediation between a complex drawing (e.g., a character, or a notation) and its Interpretation (e.g., Gestalt, or Holistic Processing; see Fig. 1). If readability exists and its de-codification process is successful, the Complex Drawing becomes Writing, and Interpretation transforms itself to Reading. We think that by studying and analyzing the code we will understand the ease with which we read a text. Finally, the ease with which we read a written text only is possible if the codification of that mediation is completely accessible to the reader.

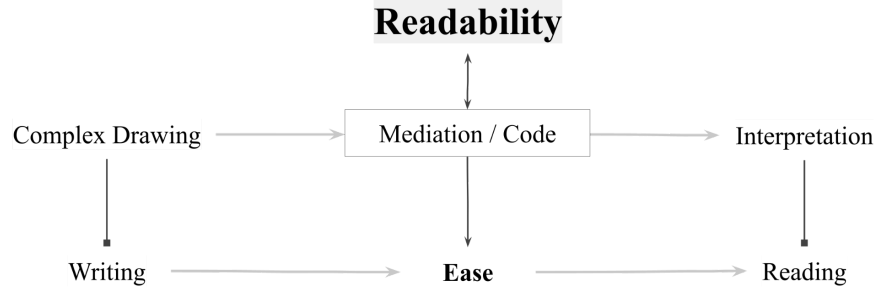


Fig. 1 Readability conceptualisation diagram. For the purpose of the quantification of Readability, we need to understand it as a mediation or a bridge between a Complex Drawing and its Interpretation.

1.1 Readability formulas

Quantification of readability is not something new. Many researchers in the 20th century created formulas that provide information about readability in literary texts (Bailin and Grafstein 2016; Benjamin 2012; Tarasov et al. 2015). One example is the *Flesch Reading Ease* formula from 1948 by Rudolph Flesch:

$$RE = 206.835 - (1.015 * ASL) - (84.6 * ASW), \quad (1)$$

where the RE stands for Readability Ease, ASL for Average Sentence Length, and ASW for Average Syllables per Word. The result is a number between 0 and 100. The higher the number, the easier it is to read the text. We need to note that the purpose of the formula is to evaluate the difficulty of texts for basic education: Texts with an RE between 90 and 100 are considered for 5th grade, and with an RE between 60 and 80 for 8th and 9th grades, etc.

The construction of readability metrics is an ongoing process. ATOS (introduced in 2000) measures book readability, while CohMetrics (developed in 2003) evaluates

psycholinguistics. Additionally, there are several revisions of earlier formulas. Measures like the *Lexile Analyzer* are standard for evaluating the relationship between the complexity of a text (vocabulary and sentence structure) and its difficulty to read. Other measures, like the *Lookahead Information Gain* (LIG), have a different approach and evaluate the amount of information that a reader is receiving (Aurnhammer and Frank 2019). What is common in almost all of these formulas, is the use of statistical tools to generate a unique quantitative value that assess readability.

Beyond educational studies, readability is a useful characterization for all purposes of texts. These formulas are also used in pharmaceutical brochures (Ravesloot et al. 2017), military equipment (Sancho Guinda 2002), and recently in online content positioning, a key aspect of Search Engine Optimization.¹

1.2 A complexity model for the quantification of music readability

Thus far, a lot of research on readability is done based on intuition that connects complexity with difficulty. Galera-Núñez (2010: 104) defines difficulty in her research stimuli by taking music scores from successive grades in two different levels of a music education curriculum. Janurik et al. (2022: 6) uses former research stimuli to replicate results. Gudmundsdottir (2010: 3) understands that the key signature (the number of sharps or flats that are applied to the rest of the pitch information in the score), and the length of the passage (the number of measures) as key factors in assessing the difficulty of the stimuli. Sheridan and Kleinsmith (2021: 4) use the amount of ink, measured with a Likert type of value, for the same purpose. Fan et al.

¹ See <https://developers.google.com/search/docs/fundamentals/seo-starter-guide> (accessed: 2022-01-01) for more details.

(2022: 5) identify and count key elements, like notes and rests, to evaluate the difficulty of written music. Endestad et al. (2020: 6) use a Likert-type assessment on a case study on three items: ‘Technical difficulty’, ‘Expression’, and ‘Harmonic tension’, all evaluated in each measure (although, they did not report why they used this segmentation). These studies, which do not lack rigor, would benefit from having a quantitative value of readability. This could go along with their qualitative understanding of the difficulty of written music stimuli. Quantification of readability could serve many music educational purposes, as readability formulas in texts do. We can improve, or at least make decisions on music writing stimuli more homogeneously; reducing subjectivity.

Although there is almost no research on the quantification of music readability, there are at least three research results that are worth mentioning: 1. *Musiplectics* by Holder (2015), that analyzes music complexity based on clarinet scores, with a musicology aim; 2. The Song and Lee model (2016), where they make an intuitive quantification of musical elements in the score, and 3. The *Score Analyzer* system by Sébastien et al. (2012), similar to the previous, but focused on piano pieces, and using seven features in a Principal Component Analysis to assess the complexity of a musical score. Based on the scarce state of the art, this tool needs to be designed as a model built on mathematical concepts such as Information Theory (see section 2.3), and computational algorithms (see 3.1). But as we see, the evaluation of the data without qualitative assessments, results in a measure of Complexity that is not directly equivalent to Difficulty. Regarding readability, Complexity is a stable assessment of relationships between states of information, while Difficulty is a dynamic concept that refers to a subjective view of our relationship with the written music. Therefore, Complexity is a static dimension, while Difficulty is a dynamic one. Current measures

on readability (as seen on 1.1) present Complexity assessed before the reading process begins. This allows readers to be less biased.

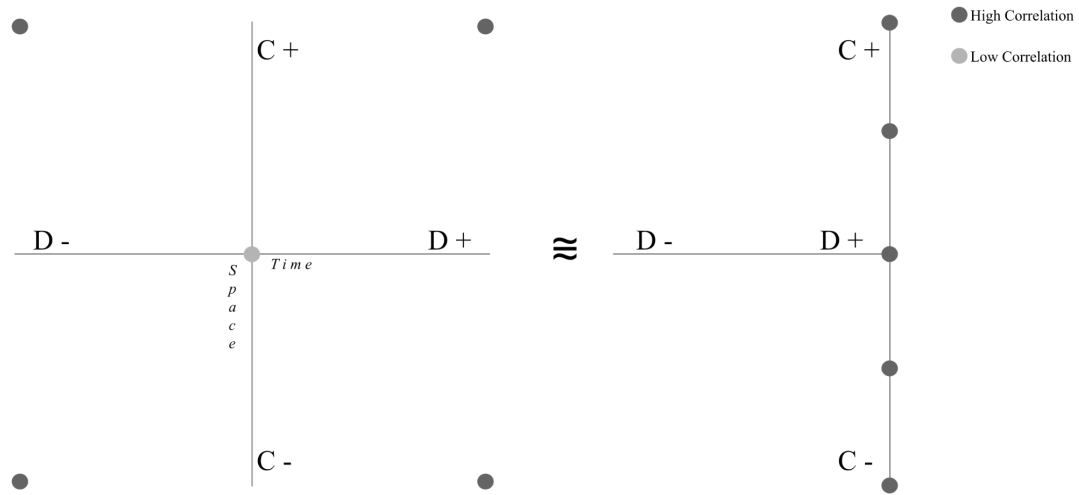


Fig. 2 Diagram of the relationship between complexity and difficulty. (Adapted from Hattie 2010). We observe two interrelated dimensions: Complexity and Difficulty. In the first one, we see an orthogonal relationship where the correlation between them is seen in the foremost regions, with a low correlation in the center. In the second relation, we see the way Readability measures are intended to be presented.

As we see in fig. 2, literature on the combination of complexity and difficulty reports that these two dimensions only have a significant correlation in the foremost regions (i.e., high complexity and high difficulty, low complexity and low difficulty, low complexity and high difficulty, and high complexity and low difficulty). Their relationship is less significant in the central region, and poor or nonexistent in the rest of the area (Aurhammer and Frank 2019; Alexandre et al. 2017; Sinha et al. 2019). This makes it almost impossible to draw a continuous and reliable correlation between significant points.²

² Like the decibel scale when describing audio intensity.

The aforementioned formulas assess syntactic conditions before the reading process begins, resulting in improved correlation outcomes. This responds to the fact that difficulty decreases over time as we perform several readings of the text; as seen on fig. 2. To attend to these issues, we will diverge from traditional evaluations of complexity in music scores (as the ones in Angeler 2020; Holder et al. 2015; Lopes and Tenreiro 2019; Menke et al. 2021; Sheridan and Kleinsmith 2021; Pease et al. 2018), and build a model that attends to how complexity and cognition research relates. We must think of the RIM as a complexity evaluation that alerts us about the information content that we have in front of us. Having information about a score beforehand allows us to estimate (guided by our expertise) the amount of time and effort that is going to take us to go from a rigid reading to a fluid one. Later, as we read the score over and over, along with adaptation, correction, and all activities involved in music reading, we become independent of the writing—we even gain perceptual amplitude (Burman and Booth 2009). The iteration process of reading progressively moves us away from the text, allowing us to build an individual, authentic performance of the music already read.

2 The RIM

In this section we provide a thorough description of the RIM. Starting from its formulation, and ending with two examples that show its behaviour, we report its conceptualization and applicability.

The necessity for a quantitative analysis of readability led us to the creation of a metric that can provide rigorous data about the syntactic complexity of written music. Based on readability formulas, and reports on music reading research we notice the opportunity to design a tool that helps musicians to assess difficulty features in the

score. At this point, the RIM is the result of applying an algorithm to musical data to obtain a series of indicators, which we intend to operationalize to design an index.³ As this model intends to predict how a musician understands readability, the index that we present is built in the line of Cognitive Musicology (Laske 1988).

2.1 Ecological considerations

As inferred from above, Complexity measurements do not consider the way written music is read. Musicians, professional or not, never read the whole score at once (at one glance), they naturally segment their access. This is clear when measuring the location of eye fixations in relation to eye saccades when reading music (Viljoen and Foxtrot 2020). Sheridan and Kleinsmith (2021) describe an intuitive way of measuring how big the focus point of each glance is. They use three segmentations of the 20/20 gaze: Foveal (as the most accurate view of a music notation fragment), Parafoveal, and Perifoveal. Also, Chitalkina et al. (2020) observed pupil size when reading music, a line of work that can be related to cognitive workload; as in the Index of Cognitive Activity (ICA) (Marshall 2002). The RIM's model segments written music into windows that mimic the way a musician naturally reads its content, considering music reading as an active process. After the segmentation is done, similarly to other models, we assign an entropy value (eq. 3) to the information contained in the window. However, before adding all values together, we perform a 'familiarity' assessment between the windows, using the Kullback-Leibler (KL) divergence score (eq. 4). The

³ Operationalization is a verb used in statistical jargon to put together all strategies that interrelate variables. When an index is the goal, a series of indicators is obtained, and a decision is made about which is the best way to interrelate them to obtain a single value, as in the *Flesh Reading Ease*'s equation (Eq. 1).

latter follows the literature on the relationship between Cognition and Information Theory research.

To model this windowing or segmenting of the score in glances, we use the notion of ‘tactus’ (Lerdahl and Jackendoff 1983; Malbrán 2007) for partitioning the score into units of foveal focus —a quantization unit of reading. This value divides the measure into a number of hierarchical rhythmic levels, but we will only consider the first three: Pulse, Accent, and Whole Measure. In this text, we create the notion of Music Reading Unit (MRU) to refer to each window. With this concept, we integrate analogies like ‘chunks’ (Sheridan and Kleinsmith 2021), ‘patterns’ (Viljoen and Foxtrot 2020), or ‘holons’ (Angeler 2020), and relate them to bigger structures used to represent long-term memory —like ‘templates’ (Sheridan and Kleinsmith 2021).⁴ Following some ideas from Gestalt (Eden Ünlü and Ece 2019), Holistic Processing (Viljoen and Foxtrot 2020), and Computational Vision (Arnheim 2020), we know that the boundaries of maximum information that one can absorb in each stimulus, reported in the literature, oscillate between two pulses (Stenberg and Cross 2019) and four elements (Baddeley 2003), or between one and seven notes (Mills and McPherson 2015).

⁴ This form of working memory traces back to research in chess players (see Gobet and Simon 1996).

A

$\text{♩} = 80$

mp cresc. *mf decresc.*

Inferior limit = 1/4: Tactus pulse level
Superior limit = 4/4: Tactus whole measure level

B

Inferior limit = 1/16: Tactus pulse sublevel (binary)
Superior limit = 1/8: Tactus pulse level

t1 t2 t3 MRU

Fig. 3 Two examples of MRU segmentation. First in the lullaby *Ah vous dirai-je, Maman*, and then in measures 26-29 from the *Largo* of *Piano Concerto No. 3* by Beethoven. Note: Tactus levels are t1 = pulse; t2 = accent, and t3 = full measure. The Beethoven's fragment is an original transcription from Dover 1983, (the author's version was made with *MuseScore 3*).

In addition, we need to consider that musical gesture is composed of several types of musical notation that mathematicians call Classes (but musicians normally refer to them as Elements, Dimensions, or Parameters), working at the same time (Lepper et al. 2019; Prince et al. 2009; Viljoen and Foxcroft 2020). Thus, we will measure all pertinent notational classes in the score sequentially, and assign a weighted value for each one (i.e., a statistical factor that distinguishes them from one another). As discussed below, this method brings readability formulas to the field of music notation, as most of them assign the same weight to all characters.

2.2 Information Theory and Cognition

Like most readability formulas, we will take advantage of Information Theory (IT) to perform calculations. IT is an extensive field that emerged in the middle of the 1940s

for measuring information, mostly in computer systems. Psychology, along with many disciplines including arts, took advantage of its measures (e.g., Shannon's Entropy) with fruitful results. However, it is necessary to say that this relationship has not always been good (Alexandre et al. 2017; Aurnhammer and Frank 2017; Sayood 2018; Thornton 2013). A serious disagreement between IT and psychology emerged in the late 1960s, when researchers began to understand that those measurements do not reveal exactly how cognition works, or how the brain processes information. After demonstrating that several psychological laws that connected IT and psychology were poorly conceived, cognition specialists began to achieve a better understanding of how information is embodied in humans. Experiments on sensing how neurons transmit electricity using the dual process of polarization–depolarization inform us that their capacity to send information relies not only on the amplitude of the message, but also on the state of the organism at one specific point (Candadai 2021; Fan 2014; Sprevak 2020).⁵ This means that embodied measurements today do not offer information about the mass of the message, as we could guess when applied to computers, but a statistical approximation of the activity in the brain. Current research suggests a two-way measure including the 'Vehicle' of information —e.g., entropy— and the 'Environmental State' —including the previous measure for prediction renewal, e.g., KL divergence— (Sprevak 2020). These ideas suggest the necessity of serious considerations in IT evaluations when applied to cognition research.

⁵ In these cases, IT measurements like entropy focus on measuring the average time in which this polarization–depolarization occurs in groups of neurons, and this allows us to know the capacity that the group has for sending information.

2.3 Selection of Written icons

For this version of the RIM, we will use Common Western Music Notation (CWMN),⁶ present in most cognitive approaches to written music (Angeler 2020; Burman and Booth 2009; Chase 2006; Eden Ünlü and Ece 2019; Holder et al. 2015; Kurkela 1989; Mills and McPherson 2015; Sheridan and Kleinsmith 2021; Slevc and Okada 2015; Stenberg and Cross 2019; Viljoen and Foxcroft 2020). This type of notation has a number of classes that have a different impact on the musical gesture. E.g., a musician intuitively knows that it is not the same to read a staff line, a slur element, or an ornamental one. Given that the readability formulas assign the same relevance to each alphanumeric value, we need to distinguish manually the notational classes, selected from CWMN. This selection was made beforehand with the aid of a simple statistical analysis on books of music education that teach music notation. With this strategy, we can identify a subset of the most used CWMN in music teaching, composition, and research. According to their Position (the order in which the specific notation is taught) and Preference (the percentage of appearance of a specific notation in books), the notational elements were filtered and weighted, resulting in a value that distinguishes each selected notation element.⁷

⁶ This terminology is standard in the field, and it refers to the notation used in the Common Practice period in Western Music. It is related to tonal music, and it is taught in all major music education institutions (Gudmundsdottir 2010).

⁷ The full text explaining this in detail is in print.

Classes implemented	Weight
Pitch	0.122
Clef	0.134
Rhythm Figure	0.136
Rest	0.156
Barline	0.162
Augmentation Dot	0.175
Tie	0.182
Accidentals	0.187
Rhythm Delta Interval	0.191
Pitch Delta Interval	0.191
8ve. Sign	0.171
Fermata	0.196
Dynamic	0.217
Slur	0.210
Wedge	0.185
Articulation	0.213
Repetition Sign	0.213
Agogic	0.208
Ornament	0.214

Table 1: Weighted factor obtained through entropy measurements for each notation class that we will use for classification purposes. Note: The number next to each notation class corresponds to its statistical relationship between the order in which it is taught, and the number of times that it appears on music education books.

3 Description of the measurements

Alongside cognition theories on music reading, the RIM is built using IT to quantify the information contained in messages, isolating information from noise. As we mentioned, the readability model that we built is based on recommendations from IT and Cognitive Studies. We know that to save cognitive load, brain processes depend on the continuous extraction of patterns in the environment, thus allowing the possibility of making predictions about future events (Friston 2017). Perceptual

processes are increasingly understood as active processes in which the brain creates generative models of the environment to predict incoming stimuli (Mencke et al. 2021). Therefore, we understand that there is information present in the score (environment–neuron relationship), but there is also an involuntary prediction of its content (neuron–environment). We will assign a value to music symbols (measuring the statistical relationship between density and diversity) as the vehicle of information with Shannon’s entropy calculations. Also, we consider its current environmental state with the aid of the KL divergence. The latter is adapted from the concepts of Sprevak (2020) in terms of addressing the relationship between two different pieces of information: Inference and representation, which is understood as an additive relationship between information and environmental states.

3.1 The algorithm

The functionality of the RIM starts with an algorithm: For a sample score, we import the notational classes of CWMN with their weighted values and perform a Music Information Retrieval (MIR) strategy for acquiring each one individually. Then, we make a histogram that includes all occurrences in each subclass of the score (e.g., a histogram of pitches of the whole pitch information). This histogram is converted into probability tables (i.e., frequency to probability distribution) following the equation

$$p(x_i) = x_i / X, \quad (2)$$

where $p(x_i)$ is the probability of a specific sub–element.

With this information, we make a histogram for each MRU in each specific class. However, we will not build probability distributions for these. Instead, the histograms

themselves will be sufficient to assign a quantitative value, according to the overall probabilities measured before. So, with the data obtained, we perform two operations:

1. An Entropy measure of the MRU density and diversity with Shannon's equation:

$$H(X) = -\sum_{i,n} p(x_i) * \log_2(x_n), \quad (3)$$

where the sum of all the probabilities is multiplied by their base 2 logarithm, which results in the amount of information combined in each segment —i.e., the information vehicle in each MRU.

2. A KL divergence, which measures the surprise or familiarity between the current MRU and the previous one. This strategy is common when measuring Cognitive Control (Alexandre et al. 2017; Aurnhammer and Frank 2019) and is used to evaluate the relationship between the initial beliefs or expectations about a stimulus (the previous MRU) and the updated belief after the new stimulus arrives (the current MRU). The equation is as follows:

$$D_{kl}(P||Q) = \sum_{i,n} p_i * \log_2(p_i/q_i) + ... + p_n * \log_2(p_n/q_n), \quad (4)$$

where P is the expected notational combination (the same as the previous probability table), and Q, the current. This calculation represents the environmental state of information.

Alongside these calculations, we evaluate the number of black pixels in each measure. After an overall summation of both entropy and divergence for each MRU, we get the information needed to calculate the indicators in the score.

Lastly, readability formulas do not consider if the text is underlined, in *italics*, **bold**, (between parentheses) or ‘commas’, and obliterating these factors in CWMN is a mistake. To attend this, we developed two final strategies to build the algorithm with syntactic rigor: 1. We split information into two groups: a. Explicit information, that is, the notation that is always present, like notes, pitches, or dots; and b. Implicit information, that is, the notation that appears only once and affects all subsequent gestures, such as dynamics and expression. To measure this contrast, we adapted the tacit-explicit converter exposed in Sudhindra et al. (2017), assuming that explicit and implicit information are parallel (both are knowledge). This allowed us to treat both types of information as equal—at least in weight. The explicit notation will be measured with its original subclass name, and we will add a T (as in Tacit) to the implicit information —e.g., a subset of dynamics information in a measure with four quarters could be [**mf**, **mfT**, **mfT**, **mfT**], and 2. We will take apart a set of notation classes: Metronome Mark, Time and Key Signatures, Character, *Tempo*, and Clef. These classes will function as structural and global information, grouped into a header information, similar to the header in a MIDI file. We report that, even though the implications of *tempo* information do not imply fixed decision, the information denotes a value, and this is what the musician initially reads. If the performer does not play in *tempo* or any other indication, it does not mean he has not read it.

After all calculations are done, the algorithm returns five values as indicators of the relationship between complexity and difficulty:

1. MRUs: The extension of the music fragment modulated by a tactus level. The number of MRUs is associated with how big the fragment to be read is (i.e., the number of systems or pages in a printed score), and how much Cognitive

Load is necessary to read the whole fragment. As we mentioned before, Gudmundsdottir (2010) uses this feature as an indicator of difficulty (see Fig. 3).

2. Ink Amount: This value is taken from Sheridan and Kleinsmith's inference on difficulty for scores (2020). It indicates the approximate amount of work that the eye must do to capture an MRU. To obtain this value, we count the non-white pixels in the digitized image of the score.⁸

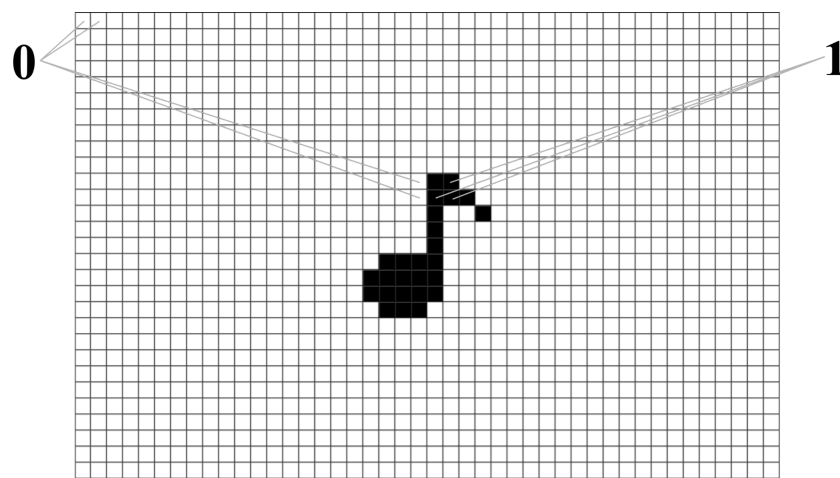


Fig. 4 Diagram for the Ink Amount indicator. Note: '0' represents pixels with no visual relevance, and '1' represents all non-white pixels, or information present in the score.

3. Fifths: The average change in key signatures in the written fragment, calculated with the Header Information. It is a measure of the area under the curve of the number of sharp or flat symbols in Key signatures throughout the score.

⁸ At the moment of writing this text, we found the average of pixels in the whole measure, and not in the tactus window.

$$\sum_{i=1}^N x_i / N, \quad (5)$$

where x_i is the number of accidentals due to key changes in the i th measure, and N is the total number of measures in the fragment. Gudmundsdottir (2010) also uses this information for the assessment of difficulty.



Fig. 5 Diagram for the Fifths indicator. Note: Following eq. 5, in example A, 0.5 is the relationship of half of the fragment with or without an accidental, and the number of them. In example B, three quarters of the fragment have one sharp.

4. Bits per MRU: The isolated amount of information in each MRU is calculated without Header Information modulation. It is a sort of analytical view of the score that does not relate to the speed with which we need to read the notation. Fan et al. (2021) use this indicator.

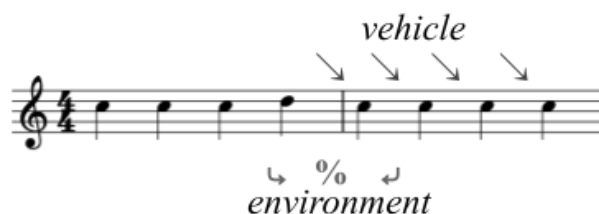


Fig. 6 Diagram of the Bits per MRU indicator.

5. Free Energy: The total information of the score, modulated by the Header Information. This indicator translates Friston's (2017) Free Energy principle, as the need to store energy to adapt our predictions about the environment. This value is similar to the computer's 'bits per second' information, and it reflects a performing view of the score, according to the speed with which we need to read the information. We could have used the term Channel Capacity as it is normal in this scenario, but it is understood as the maximum amount of information that will go through a channel in a computer, and as we saw, it is better to understand entropy values as activity and not as a mass. We think that the least amount of energy available is more accurate for this value.⁹

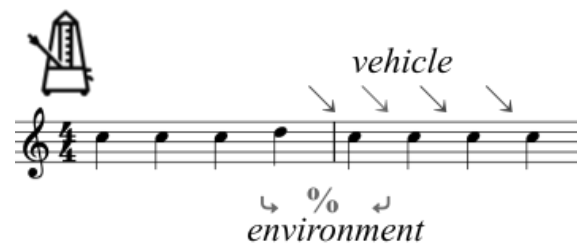


Fig. 7 Diagram of the Free Energy indicator. Note: As in the previous indicator, the evaluation of density and diversity of classes, related to its temporal appearance is the same.

Finally, the diagram of the algorithm shows all procedures and highlights the indicators during the process.

⁹ This concept is like Zipf's Law of Least Effort (Buchanan 2016), or Rayner and Pollatsek's Minimal Attachment Principle (Nordquist 2020). As mentioned before, it is common to assume that human beings tend to save mental energy to make other things while reading, such as tuning and making actual music.

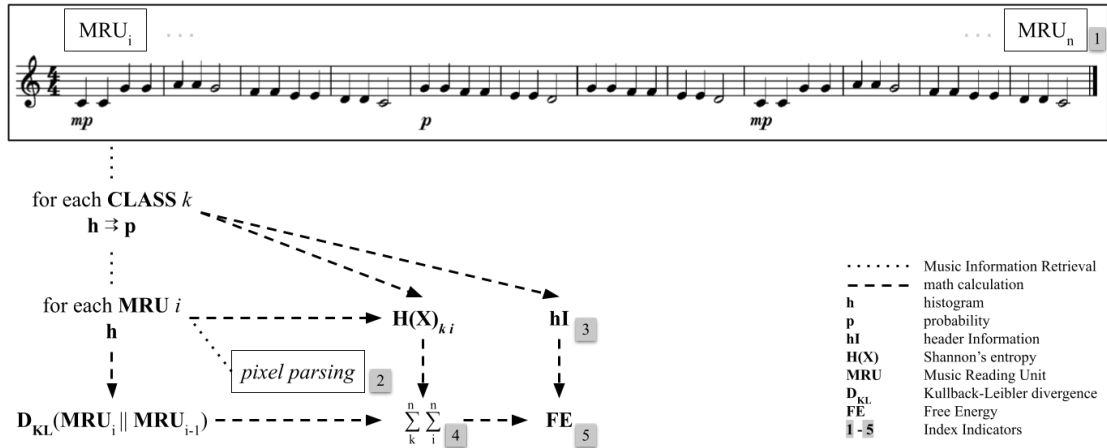


Fig. 8 Diagram of the RIM's algorithm. Note: Gray numbers show, sequentially, how Indicators are built with the MIR strategy. Below we describe a basic example of its operation.

3.2 Example 1

We will exemplify the application of the algorithm using the first measures of the lullaby *Ah vous dirai-je, Maman*.



Fig. 9: Transcription of the theme *Ah vous dirai-je, Maman*. (Transcription made with *MuseScore 3*).

STAGE	ACTION	VALUES
For each CLASS (e.g., Pitch)	<i>Histogram</i>	C = 6/42, G = 10/42, A = 4/42, F = 8/42, E = 8/42, D = 6/42
	<i>Probability table</i>	C = 0.143, G = 0.238, A = 0.096, F = 0.19, E = 0.19, D = 0.143

For each MRU (e.g., tactus level 3 = Pulse)	<i>Histogram</i>	1 C, 2 C, 3 G, 4 G, 5 A, 6 A, 7 G, 8 null, 9 F, 10 F, 11 E, 12 E, 13 D, 14 D, 15 C, 16 null, 17 G, 18 G, 19 F, 20 F, 21 E, 22 E, 23 D, 24 null, 25 G, 26 G, 27 F, 28 F, 29 E, 30 E, 31 D, 32 null, 33 C, 34 C, 35 G, 36 G, 37 A, 38 A, 39 G, 40 null, 41 F, 42 F, 43 E, 44 E, 45 D, 46 D, 47 C, 48 null. => 48 MRUs
For each MRU in each CLASS	<i>Entropy</i>	1 0.401, 2 0.401, 3 0.493, 4 0.493, 5 0.325, 6 0.325, 7 0.493, 8 0, 9 0.455, 10 0.455, 11 0.455, 12 0.455, 13 0.401, 14 0.401, 15 0.401, 16 0, 17 0.493, 18 0.493, 19 0.455, 20 0.455, 21 0.455, 22 0.455, 23 0.401, 24 0, 25 0.493, 26 0.493, 27 0.455, 28 0.455, 29 0.455, 30 0.455, 31 0.401, 32 0, 33 0.401, 34 0.401, 35 0.493, 36 0.493, 37 0.325, 38 0.325, 39 0.493, 40 0, 41 0.455, 42 0.455, 43 0.455, 44 0.455, 45 0.401, 46 0.401, 47 0.401, 48 0
	<i>KL divergence</i>	1 0.063, 2 0, 3 0.143, 4 0, 5 0.143, 6 0, 7 0.143, 8 0.056, 9 0.063, 10 0, 11 0.143, 12 0, 13 0.143, 14 0, 15 0.143, 16 0.056, 17 0.063, 18 0, 19 0.143, 20 0, 21 0.143, 22 0, 23 0.143, 24 0.056, 25 0.063, 26 0, 27 0.143, 28 0, 29 0.143, 30 0, 31 0.143, 32 0.056, 33 0.063, 34 0, 35 0.143, 36 0, 37 0.143, 38 0, 39 0.143, 40 0.056, 41 0.063, 42 0, 43 0.143, 44 0, 45 0.143, 46 0, 47 0.143, 48 0.056
For each MRU in the score	<i>Sum</i>	1 0.464, 2 0.401, 3 0.636, 4 0.493, 5 0.468, 6 0.325, 7 0.636, 8 0.056, 9 0.518, 10 0.455, 11 0.598, 12 0.455, 13 0.544, 14 0.401, 15 0.544, 16 0.056, 17 0.556, 18 0.493, 19 0.598, 20 0.455, 21 0.598, 22 0.455, 23 0.544, 24 0.056, 25 0.556, 26 0.493, 27 0.598, 28 0.455, 29 0.598, 30 0.455, 31 0.544, 32 0.056, 33 0.464, 34 0.401, 35 0.636, 36 0.493, 37 0.468, 38 0.325, 39 0.636, 40 0.056, 41 0.518, 42 0.455, 43 0.598, 44 0.455, 45 0.544, 46 0.401, 47 0.544, 48 0.056
1 MRUs	<i>Sum</i>	48
2 Ink Amount	<i>Average</i>	7,357
3 Fifths	<i>Area under the curve</i>	0
4 Bits per MRU	<i>Average</i>	0.45
5 Free Energy (e.g., J= 80, 750 ms. per Pulse)	<i>Weighted average</i>	0.6

Table 2: Stages, actions, and values that result from the implementation of the RIM in the current score. In the case where there is none information available, as when pitch information is not present in the second tactus level (pulse) in a half rhythmic notation, we put a *null* value. There is a provisional implementation of the algorithm in JavaScript, available at <https://github.com/patricio1979/sComplexity> (accessed: 2023-01-01).

3.3 Example 2

The previous example shows five indicators of written music complexity in a small musical fragment. However, these values alone do not express much of the written content in the score. They are meaningful when they are compared.

In ‘doubling’, a common technique in music editorial design, we can see differences in the values of the indicators. As a case study, we will consider the measures 26-29 from the *Largo* of the *Piano Concerto No. 3* by Beethoven (Dover 1983), and compare them with its ‘doubled’ version.



Fig. 10 Measures 26-29 from Beethoven’s *Largo* of the *Piano Concerto No. 3*. (Original transcription from Dover (1983) on *MuseScore 3*).

For the original fragment, the algorithm obtains the following values: MRUs = 12; Ink Amount = 28,958.5; Fifths = 4; Bits per MRU = 23.71, and Free Energy = 43.109.



Fig. 11 The same fragment, but with the ‘doubling’ technique applied to it.

After we double the rhythms in the fragment, the algorithm gives these values: MRUs = 12; Ink Amount = 25,521.75; Fifths = 4; Bits per MRU = 23.71, and Free Energy = 43.109.

The comparison exposes that doubling affects only Ink Amount, but not the other indicators. This reveals that the indicators are not fully interdependent, as usually happens in statistical results when analyzing art with IT measurements. This effect, called ‘burstiness’ (Ogura 2013), shows the abnormal (also called parametric) dispersion of information throughout the text¹⁰ –in our case, occurrences in music notation classes. It can also be seen in the behavior of these indicators in specific situations. E.g., a score can show changes in Fifths, but not in Free Energy. Likewise, a change in the number of MRUs does not necessarily relate to its Bits per MRU. However, we note that Bits per MRU and Free Energy do show a relationship in example 2, but this is not always true: If we have two equal scores with different *tempo*, they will show the same value for bits per second, but different values for Free Energy.¹¹

¹⁰ Abnormal in the sense that statistical analysis when entropy measures dispersion in gas molecules is normally distributed in a space, while entropy used in literature with IT appears on arbitrary locations such as rhymes, vowels in poetry, grammar, orthography, etc.

¹¹ It is natural to think that, with more research, it will be possible to give some sort of correlation value between the indicators.

4 Discussion

Taking into account the recommendations for building connections between Cognition Studies and IT, we built five indicators knowing how musicians direct their gaze on the score. The amount of information modelled and the segmentation of the score take into account current reports in psychology.

Unlike related research, the RIM is built knowing how musicians acquire musical data when reading; this is, taking into account the number of glances that they make on the score. Contrary to the current complexity analysis, where the object (score) is assessed in its total extension, our approach has a human perspective on complexity, and its ecological concern can be exploited in pedagogical planning –as readability formulas do with literature analysis. The RIM's indicators reflect difficulty features reported in music cognition and reading research. Besides the improvement of readability decisions in research, as the ones of Galera-Núñez (2010), Gudmundsdottir (2010), Fan et al. (2022), Endestad et al. (2020), we think that complexity measures in music (Lopes and Tenreiro 2019; Pease and West 2018), mathematical approaches to MIR (González-Espinoza et al. 2020; Lepper et al. 2019; Prince et al. 2009), and other areas can take advantage of tools like the RIM. This approach will embody their complexity measures, and consider how humans perceive information.

Although the index models a visual to verbal transcodification (Janurik et al. 2022), there is an open field for research in the other two forms of music reading (auditory, and performative). The RIM only focuses on musical data that can be digitized, but as we saw, the five indicators attend to most of the difficulty features used in music cognition research. With this Cognitive Musicology Model, we can approach Music Cognition research on Readability in a better way than before.

As we saw in the second example (Beethoven's case study) the indicators are not directly correlated, so we will need a factor to multiply each indicator, as we saw in the *Flesch Reading Ease* formula (eq. 1). Perhaps a Machine Learning algorithm can be the best solution, but we need to do further research. Also, this case study shows that the RIM's indicators are useful for observing the impact of current editorial criteria on readability. It confirms as well that the indicators' statistical description show parametric and not normal data distributions, as it occurs in statistical analysis of the arts.

4.1 Index operationalization

At this point, it is still soon to operationalize the indicators to build a general RIM. Entropy does not reflect difficulty in the score; it only describes the relationship between diversity and density of written notational information, i.e., its intrinsic complexity. It is necessary to do further experimentation, e.g., with Implicit Cognitive Load (Galera-Núñez 2010: 74-75), but also measuring Cognitive or Executive Control with IT strategies (Aurnhammer and Frank 2019; Fan 2014; Gao et al. 2021; Mencke et al. 2021; Slevc and Okada 2015; Thornton 2013; Viljoen and Foxcroft 2020). To do this, we need to use well-suited technology to calculate the most approximated connection between the Complexity measured with the RIM's indicators, and the Difficulty, reported as a mental activity in music performers. If a correlation between brain activity and the indicators is found, we expect that a factor can be applied to each one, to operationalize them. Probably, the main issue is that our entropy measures are, as expected, noiseless, and having this type of isolation in brain sensing is almost impossible now.

4.2 Limitations

The RIM's calculations are made considering the latest research on music reading, but as Madell and Hébert report (2008), this area grows slowly and relies heavily on literature research. Features like surprise, measured by KL divergence, are intermediate solutions.

At this point,

- We only use CWMN, but we know that there are several other options for performing readability assessments. It could be interesting to adapt the RIM to other types of notation and compare them on, e.g., the same score written with different symbols –as it often occurs when new notation is proposed.
- We do not include idiomatic notation (e.g. bowing for strings, flute fingering positions, guitar or piano fingerings, pedal diagrams for harp, organ etc.). This would lead to us to build a number of RIM measurements, modified for each instrument.
- We do not weigh internal objects in classes, (e.g., rhythmic indications for crochets, or quavers are treated with the same value). Without specific research on how those elements, and their changes affect brain activity, we settle with a density and diversity value provided by entropy measurements.

In computational terms, as we use a rich file format (musicXML) for MIR, there is not a big corpus to experiment with.

Last, as we mentioned, we need extensive research on human behaviour, and its quantitative relationship to the RIM's indicators. This point is currently being addressed.

5 Conclusions

Even Though the RIM is in its early stages, we designed a tool that analyses readability using tools from Music Cognition research and IT. At this stage, the indicators that result from the algorithm successfully measure the complexity of a score written with CWMN. The two examples provided show that these indicators behave as are reported in the literature.

5.1 Educational implications

The RIM is a tool that analyses a fragment of a score, giving five features of complexity that can be included as a readability feature. This new quantitative measurement is also useful in music education. Institutions can now compare the ease with which scores are read, being capable of: 1. Homologate complexity in two or more scores in a repertoire; 2. Consistently build a progression of score difficulty, correcting a previous one or designing a new one; 3. The mental load of the students, their performance with the instrument, their score level, and many other features can now have quantitative support; 4. Two scores from different styles could have the same complexity, bringing new music more easily into the curricula; 5. Given the scarce literature on the systematization of a new institutional syllabus, the RIM's indicators can help in managing subjectivity in an assertive way; 6. Last, this tool, as we said, in its current stage can go deeper into the mind of the musician, assisting the

development of readability criteria and fasten several processes in the music institution.

5.2 Future developments

To use another music notation, we will need to make fundamental changes. The windowing design will also need to be modified, as the weighting factor of each new character.

We can also expand our written music range using the same strategy as we did here. E.g., if we want to include 20th. century notation in the curriculum of music teaching planning, we need to collect all possible references to the notation used in one instrument or composer, and build another set of weights. We could perform the same algorithm and expect similar results with this information.

A useful feature that can help focus groups of music teachers, is the possibility of customizing the RIM's algorithm for, e.g., not including any indicator, or giving it less weight. We hypothesize that the best way to operationalize the indicators is to design another algorithm that can take a corpus of scores, alongside with a user's criteria, in order to provide a single value that assesses readability in that specific corpus. Another feature planned has to do with music style and familiarity. This can be useful when comparing scores with the intention of interchanging or substituting them in a music curriculum.

We plan a series of experiments that empirically test the RIM. This is a challenge due to several aspects, mostly concerning the sample population available. However, as implied before, it could be helpful to use well-suited electrophysiological equipment

to isolate the Regions of Interest (ROIs) in the brain related to readability (as in Aurnhammer and Frank 2019 and Sinha et al. 2019).

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7 Competing interests

The authors have no competing interests to declare.

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